

ECEN 662: Estimation & Detection Theory

Lecture 10: Decision-Theoretic Framework

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- The *frequentist risk* associated with a decision rule δ is defined as

$$R_\delta(\theta) := \mathbb{E}_\theta[L(\theta, \delta(X))]$$

where the expectation is over $X \sim f_\theta(x)$

Usual loss functions

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- Note that both the squared-error loss and the absolute-error loss are *convex* functions of d for any given θ , so both are examples of a *convex loss function*
- When the decisions have a *categorical* nature, the most usual choice of the loss function is the *0-1 loss*

$$L(\theta, d) = 1_{\{\theta \neq d\}}$$

for parameter estimation problems and

$$L(\theta, d) = 1_{\{\theta \notin \Theta_d\}}$$

for detection

Mean squared-error and bias-variance tradeoff

- Under the *squared-error* loss, the frequentist risk is given by the *mean squared-error (MSE)*, which can be further decomposed as the sum of a *bias* term and a *variance* term:

$$\begin{aligned}R_{\delta}(\theta) &= \mathbb{E}_{\theta}[(\theta - \delta(X))^2] \\&= \mathbb{E}_{\theta}[(\theta - \mathbb{E}_{\theta}[\delta(X)] + \mathbb{E}_{\theta}[\delta(X)] - \delta(X))^2] \\&= (\theta - \mathbb{E}_{\theta}[\delta(X)])^2 + \mathbb{E}_{\theta}[(\mathbb{E}_{\theta}[\delta(X)] - \delta(X))^2] \\&= (\theta - \mathbb{E}_{\theta}[\delta(X)])^2 + \text{Var}_{\theta}[\delta(X)]\end{aligned}$$

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- In short,

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- Potentially, there is a *tradeoff* between the *bias* term and the *variance* terms in achieving a smaller value of the MSE

Gaussian with unknown mean and variance

- Consider the following family of estimators

$$\delta_b(X) = bS^2$$

for the variance parameter σ^2 , where

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu})^2$$

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- By *Cochran's theorem*, we have

$$\begin{aligned} \text{Bias}^2 &= (b-1)^2 \sigma^4 \\ \text{Variance} &= \frac{2b^2 \sigma^4}{n-1} \end{aligned}$$

and hence

$$\text{MSE} = \left((b-1)^2 + \frac{2b^2}{n-1} \right) \sigma^4$$

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- In terms of the MSE, we have

$$\mathbb{E}_{\theta}[(\theta - \delta_{\frac{n-1}{n}}(X))^2] = \frac{(2n-1)\sigma^4}{n^2} < \frac{2\sigma^4}{n-1} = \mathbb{E}_{\theta}[(\theta - \delta_1(X))^2]$$

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- However, *neither* achieves the minimum MSE, which is achieved at $b = \frac{n-1}{n+1}$ with the minimum MSE given by

$$\mathbb{E}_\theta[(\theta - \delta_{\frac{n-1}{n+1}}(X))^2] = \frac{2\sigma^4}{n+1}$$

Frequentist risk as an evaluation criterion

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- One may hope for a decision rule δ that *uniformly* minimizes the risk $R_\delta(\theta)$, but such cases *rarely* occur unless the space of decision rules is *restricted*. We shall come back to the restriction idea a bit later in this lecture

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- More generally, however, we can use the frequentist risk to weed out some of the decision rules from being considered, at least from the risk-reduction point of view

Admissibility

- Formally, a decision rule δ is *inadmissible* if it is *dominated* by another decision rule δ' , i.e., for *every* $\theta \in \Theta$ we have

$$R_{\delta'}(\theta) \leq R_{\delta}(\theta)$$

and there exists at least one value $\theta_0 \in \Theta$ such that

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- For example, for Gaussian with unknown mean and variance, both the UMUVE $\delta_1 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ and the MLE $\delta_{\frac{n-1}{n}} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ are *inadmissible* with respect to the squared-error loss, as both are *dominated* by $\delta_{\frac{n-1}{n+1}} = \frac{1}{n+1} \sum_{i=1}^n (X_i - \bar{X})^2$

Risk reduction under convex loss functions

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- (*Generalized Rao-Blackwell theorem*) Let δ be a decision rule and

$$\delta'(T) = \mathbb{E}_\theta[\delta(X)|T]$$

where T is a *sufficient* statistic of θ . Then for any *convex* loss function $L(\theta, d)$, we have

$$R_{\delta'}(\theta) \leq R_\delta(\theta), \quad \forall \theta \in \Theta$$

where

$$R_\delta(\theta) = \mathbb{E}_\theta[L(\theta, \delta(X))] \quad \text{and} \quad R_{\delta'}(\theta) = \mathbb{E}_\theta[L(\theta, \delta'(T))]$$

Proof

- First note that by *Jensen's inequality*, we have

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- This immediately gives us

$$R_{\delta'}(\theta) \leq R_\delta(\theta)$$

for any $\theta \in \Theta$

Randomized decision rules

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- Formally, let Z be a random source over $\mathcal{Z}$ that is *independent* of the observation X . A *randomized* decision rule $\delta^* : \mathcal{X} \times \mathcal{Z} \rightarrow \mathcal{D}$ is a mapping that maps an observation x to a decision d based on the realization of the random source Z

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- The *risk function* that corresponds to a *randomized* decision rule δ^* is defined as

$$R_{\delta^*}(\theta) := \mathbb{E}_{\theta}[L(\theta, \delta^*(X, Z))]$$

where the expectation is over both $X \sim f_{\theta}(x)$ and Z

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- Now let $\mathcal{R}$ be the set of *all* possible risk vectors that can be achieved by any *deterministic* decision rule. Then $\mathcal{R}$, as a subset of $\mathbb{R}^K$, can take some very *arbitrary* shapes, and in general there are *no* meaningful ways to characterize such sets

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- On the other hand, let $\mathcal{R}^*$ be the set of *all* possible risk vectors that can be achieved by any *randomized* decision rule. Then, it is easy to see that $\mathcal{R}^*$ is the *convex hull* of $\mathcal{R}$ and hence must be *convex* set, which can be much easier to characterize

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- Note that $\delta'(X)$ is a *deterministic* function of X . Furthermore, by *Jensen's inequality*,

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- Two most often collapsing strategies are *minimax* and *Bayes* estimators

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- By the *generalized Rao-Blackwell* theorem, the above result holds not just for the squared-error loss, but for any *convex* loss functions